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Knowledge Neurons in Pretrained Transformers

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Abstract

Large-scale pretrained language models are surprisingly good at recalling factual knowledge presented in the training corpus (Petroni et al., 2019; Jiang et al., 2020b). In this paper, we present preliminary studies on how factual knowledge is stored in pretrained Transformers by introducing the concept of *knowledge neurons*. Specifically, we examine the fill-in-the-blank cloze task for BERT. Given a relational fact, we propose a knowledge attribution method to identify the neurons that express the fact. We find that the activation of such knowledge neurons is positively correlated to the expression of their corresponding facts. In our case studies, we attempt to leverage knowledge neurons to edit (such as update, and erase) specific factual knowledge without fine-tuning. Our results shed light on understanding the storage of knowledge within pretrained Transformers. The code is available at https://github.com/Hunter-DDM/knowledge-neurons.

1 Introduction

Large-scale pretrained Transformers (Devlin et al., 2019; Liu et al., 2019; Dong et al., 2019; Clark et al., 2020; Bao et al., 2020) are usually learned with a language modeling objective on large-scale corpora, such as Wikipedia, where exists oceans of factual knowledge. Pretrained language models naturally play as a free-text knowledge base by predicting texts (Bosselut et al., 2019). Petroni et al. (2019) and Jiang et al. (2020b) probe factual knowledge stored in pretrained language models by fill-in-the-blank cloze queries. The evaluation shows that pretrained Transformers have a strong ability to recall factual knowledge without any fine-tuning. Roberts et al. (2020) use closed-book question answering to show that the larger a model is, the more knowledge it can store. However, most previous work focuses on evaluating the overall accuracy of text-form knowledge prediction. In this paper, we attempt to look deeper into pretrained Transformers and investigate how factual knowledge is stored.

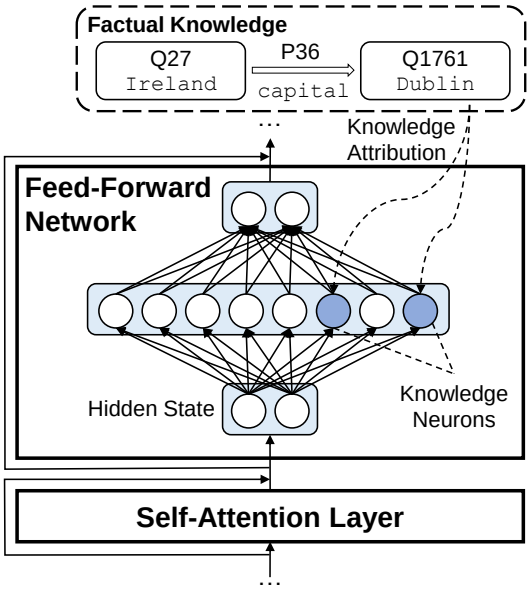

Figure 1: Through knowledge attribution, we identify knowledge neurons that express a relational fact.

As shown in Figure 1, we propose a knowledge attribution method to identify the neurons that express a relational fact, where such neurons are named *knowledge neurons*. Specifically, we view feed-forward network (i.e., two-layer perceptron) modules in Transformer as key-value memories (Geva et al., 2020). For the example in Figure 1, the hidden state is fed into the first linear layer and activates knowledge neurons; then, the second linear layer integrates the corresponding memory vectors. The key-value-memory nature (Geva et al., 2020) inspires us to propose the knowledge attribution method, which identifies knowledge neurons in feed-forward networks by computing the contribution of each neuron to the knowledge prediction.

Extensive analysis shows that the activation of the identified knowledge neurons is positively correlated to the knowledge expression, which shows

[*]Contribution during internship at Microsoft Research.

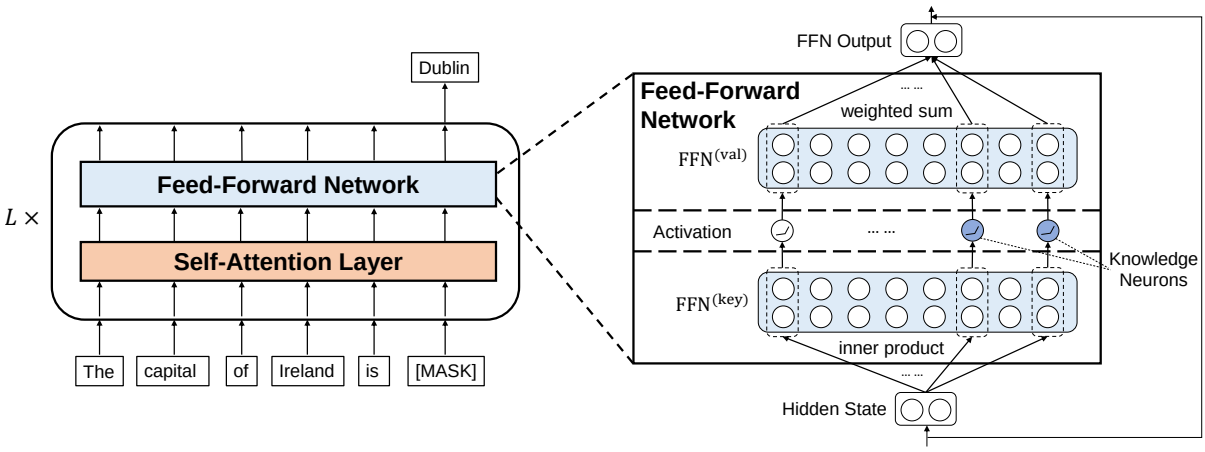

Figure 2: Illustration of how an FFN module in a Transformer block works as a key-value memory. The first linear layer $\mathrm{FFN}^{(\mathrm{key})}$ computes intermediate neurons through inner product. Taking the activation of these neurons as weights, the second linear layer $\mathrm{FFN}^{(\mathrm{val})}$ integrates value vectors through weighted sum. We hypothesize that knowledge neurons in the FFN module are responsible for expressing factual knowledge.

the effectiveness of the proposed knowledge attribution method. First, suppressing and amplifying knowledge neurons notably affects the expression of the corresponding knowledge. Second, we find that knowledge neurons of a fact tend to be activated more by corresponding knowledge-expressing prompts. Third, given the knowledge neurons of a fact, the top activating prompts retrieved from open-domain texts usually express the corresponding fact, while the bottom activating prompts do not express the correct relation.

In our case studies, we try to leverage knowledge neurons to explicitly edit factual knowledge in pretrained Transformers without any fine-tuning. We present two preliminary studies: updating facts, and erasing relations. After identifying the knowledge neurons, we perform a knowledge surgery for pretrained Transformers by directly modifying the corresponding parameters in feed-forward networks. Such surgery shows promising results, keeping a moderate influence on other knowledge.

Our contributions are summarized as follows:

- We introduce the concept of *knowledge neurons* and propose a knowledge attribution method to identify the knowledge neurons that express specific factual knowledge in the fill-in-the-blank cloze task.
- We conduct both qualitative and quantitative analysis to show that knowledge neurons are positively correlated to knowledge expression.
- We present preliminary studies of leveraging knowledge neurons to edit factual knowledge in Transformers, even without any fine-tuning.

2 Background: Transformer

Transformer (Vaswani et al., 2017) is one of the most popular and effective NLP architectures. A Transformer encoder is stacked with L identical blocks. Each Transformer block mainly contains two modules: a self-attention module, and a feed-forward network (abbreviated as FFN) module. Let $X \in \mathbb{R}^{n \times d}$ denote the input matrix, two modules can be formulated as follows:

$$Q_h = XW_h^Q, K_h = XW_h^K, V_h = XW_h^V, \quad (1)$$

$$\text{Self-Att}_h(X) = \mathrm{softmax}\left(Q_h K_h^T\right) V_h, \quad (2)$$

$$\mathrm{FFN}(H) = \mathrm{gelu}\left(HW_1\right) W_2, \quad (3)$$

where $W_h^Q, W_h^K, W_h^V, W_1, W_2$ are parameter matrices; $\text{Self-Att}_h(X)$ computes a single attention head; H, the hidden state, is given by projecting the concatenation of all heads; gelu denotes the GELU activation function (Hendrycks and Gimpel, 2016). For simplicity, we omit the scaling factor in self-attention and the bias terms.

Connections Between Self-Attention and FFN Comparing Equation (2) and Equation (3), we notice that the formula of $\mathrm{FFN}(\cdot)$ is quite similar to $\text{Self-Att}(\cdot)$, except the activation function gelu in FFN and $\mathrm{softmax}$ in self-attention. Thus, similar to the query-key-value mechanism in self-attention, it is reasonable to regard the input of the FFN as a query vector, and two linear layers of the FFN as keys and values, respectively. Similar observations are also described in (Geva et al., 2020).

3 Identifying Knowledge Neurons

Similar to (Geva et al., 2020), we view FFNs in Transformer as key-value memories as illustrated in Figure 2. We hypothesize that factual knowledge is stored in FFN memories and expressed by *knowledge neurons*. In this section, we propose a knowledge attribution method and a refining strategy to identify these knowledge neurons.

3.1 Knowledge Assessing Task

We employ the fill-in-the-blank cloze task to assess whether a pretrained model knows a fact. Following Petroni et al. (2019), each relational fact is in the form of a triplet $\langle h, r, t \rangle$, where h is the head entity, t is the tail entity, and r is the relation between them. Given a fact, pretrained models answer the cloze query x that expresses the fact but leaves the tail entity as a blank. For example, given the fact $\langle$`Ireland, capital, Dublin`$\rangle$, a possible query is "*The capital of Ireland is* __". We also call the query a *knowledge-expressing prompt*. Petroni et al. (2019) describe that a model knows a fact if it can predict the correct answer. In this paper, rather than just examining the model outputs, we identify the specific knowledge neurons that express factual knowledge.

3.2 Knowledge Attribution

Inspired by Hao et al. (2021), we propose a knowledge attribution method based on integrated gradients (Sundararajan et al., 2017). Our method can evaluate the contribution of each neuron to knowledge predictions. In this paper, we examine FFN intermediate neurons for the masked token, where the answer is predicted.

Given an input prompt x, we first define the model output $\mathrm{P}_x(\hat{w}_i^{(l)})$ as the probability of the correct answer predicted by a pretrained model:

$$\mathrm{P}_x(\hat{w}_i^{(l)}) = p(y^* | x, w_i^{(l)} = \hat{w}_i^{(l)}), \tag{4}$$

where y^* denotes the correct answer; $w_i^{(l)}$ denotes the i-th intermediate neuron in the l-th FFN; $\hat{w}_i^{(l)}$ is a given constant that $w_i^{(l)}$ is assigned to.

In order to calculate the attribution score of a neuron $\mathrm{Attr}(w_i^{(l)})$, we gradually change $w_i^{(l)}$ from 0 to its original value $\overline{w}_i^{(l)}$ calculated by the pretrained model, and meanwhile integrate the gradients:

$$\mathrm{Attr}(w_i^{(l)}) = \overline{w}_i^{(l)} \int_{\alpha=0}^{1} \frac{\partial \, \mathrm{P}_x(\alpha \overline{w}_i^{(l)})}{\partial w_i^{(l)}} d\alpha, \tag{5}$$

where $\frac{\partial \, \mathrm{P}_x(\alpha \overline{w}_i^{(l)})}{\partial w_i^{(l)}}$ calculates the gradient of the model output with regard to $w_i^{(l)}$. Intuitively, as α changes from 0 to 1, by integrating the gradients, $\mathrm{Attr}(w_i^{(l)})$ accumulates the output probability change caused by the change of $w_i^{(l)}$. If the neuron has a great influence on the expression of a fact, the gradient will be salient, which in turn has large integration values. Therefore, the attribution score can measure the contribution of the neuron $w_i^{(l)}$ to the factual expressions.

Directly calculating continuous integrals is intractable. We instead use Riemann approximation $\tilde{\mathrm{Attr}}(w_i^{(l)}) = \frac{\overline{w}_i^{(l)}}{m} \sum_{k=1}^{m} \frac{\partial \, \mathrm{P}_x(\frac{k}{m} \overline{w}_i^{(l)})}{\partial w_i^{(l)}}$, where $m = 20$ is the number of approximation steps. With the attribution algorithm, we can identify a coarse set of knowledge neurons whose attribution scores are greater than a threshold t.

3.3 Knowledge Neuron Refining

In order to identify knowledge neurons more accurately, we further propose a refining strategy. Besides "true-positive" knowledge neurons that express factual knowledge, the coarse set of knowledge neurons may contain "false-positive" knowledge neurons that express other information (e.g., syntactic or lexical information). The refining strategy aims to filter out these "false-positive" neurons.

For different prompts corresponding to the same fact, we hypothesize that they share the same set of "true-positive" knowledge neurons, since they express the same factual knowledge. Meanwhile, we hypothesize that they do not share the "false-positive" knowledge neurons as long as the prompts are diverse enough. Therefore, given multiple diverse prompts, we can refine the coarse set of knowledge neurons by retaining only neurons that are widely shared among these prompts.

Specifically, given a relational fact, the complete process to identify its knowledge neurons is described as follows: (1) produce n diverse prompts; (2) for each prompt, calculate the knowledge attribution scores of neurons; (3) for each prompt, retain the neurons with attribution scores greater than the attribution threshold t, obtaining the coarse set of knowledge neurons; (4) considering all the coarse sets together, retain the knowledge neurons shared by more than $p\%$ prompts.